

Department of Computer Science and Engineering

Academic Year 2021– 2022(Odd Semester)

Degree, Semester & Branch: III Semester B.E. ECE-‘B’ Section

Course Code & Title: EC8393 Fundamentals of Data Structures in C

Name of the Faculty member (s): Dr.V. Anusuya, ASCP/CSE

Innovative Practice Description

- **Unit / Topic: Unit 5 / Flash Cards**
- **Course Outcome:** The student will be able to appropriately choose the sorting, searching algorithms and also apply hashing concepts for a given problem.
- **Topic Learning Outcome:** TLO13.
- **Activity Chosen: Flash Cards**
- **Justification:**
 - In unit V, the students need to understand various sorting techniques such as insertion sort, bubble sort, merge sort and quick sort. Also they need to understand the technique used and complexity of each sorting technique. This activity helps the students to understand the techniques and its complexity clearly
 - **Time Allotted for the Activity:** 30 Minutes
- **Details of the Implementation:**

A flash card is a method in which the question and corresponding answer is in separate card. Question and answer of the card is prepared on sorting techniques, previously taught by the faculty in the class. Answer cards will be distributed to students in the class. Once the question will be asked by one of the students in the class from the card, student members of group in the class should identify correct answer cards within 2 minutes with their answer cards and explain the answer in front of the class for next 2 minutes. This will help the students to revise the particular topic.

After that, the students were asked to discuss the answer for the questions as shown in the figure 1,2.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PSO1
CO2	3	3	2	2	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PSO1
	3	3	2	2	2
Justification for correlation	Needs engineering and mathematical knowledge	Able to identify relevant sorting, searching	Able to interpret the complex engineering	Research-based knowledge will help to identify the suitable sorting and	able to implement different sorting and searching algorithms in IT

	for Array concept	problem for the given applications	problems with the help of the sorting algorithms	searching algorithms.	applications
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• **Images / Screenshot of the practice:**



Figure 1 Explanation of students

Screen shots



Figure 2. Explanation of students

Reflective Critique:

- ***Feedback of practice from students and other stakeholders:***
 - All the students were actively participated and enjoyed among their group members once they answered correctly.
 - Students explored the knowledge of the various sorting techniques.
 - It makes the students to gain well knowledge on different sorting and searching techniques concepts, complexity and applications.

- ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
 - Students can able to write the sorting techniques questions in Internal Test.
 - Students can able to perform different sorting techniques for any applications.
 - They can able to identify the corresponding sorting technique for the applications and implement using C programming.
 - This activity helps the students to attend the sorting technique questions in the examination.
- ❖ ***Challenges faced in implementation:***
 - Few students can get difficulties in identifying the complexity of the different sorting algorithms.
 - Some students were hesitated to express the relevant answers to the question card.

References:

- ❖ <https://www.teachingenglish.org.uk/article/using-flash-cards-young-learners>
- ❖ <https://eltexperiences.com/flashcards-in-the-classroom-ten-lesson-ideas/>

Signature of Faculty Member

HOD